

Frank Hodges

PhD Student in AI/ML

College of Engineering
Oregon State University
Corvallis, OR
☎ 440-638-9797
✉ hodgesf@oregonstate.edu
in [hodgesfm](#)

Education

- 2024–Present **Ph.D. in Artificial Intelligence**, *Oregon State University*, Corvallis, OR, 3.79
Advisor: Dr. Stephen Ramsey (expected 2027).
M.S. in Computer Science (concurrent)
- 2023–2024 **B.S. in Computer Science**, *Oregon State University*, Corvallis, OR, 3.81
- 2017–2020 **B.A. in Biology**, *The Ohio State University*, Columbus, OH, 3.12

Relevant Experience

- 2024–Present **Graduate Research Assistant — Ramsey Lab**, *Oregon State University*, Corvallis, OR
- Leading the design and implementation of an agentic biomedical RAG system (Radiant) for rare disease diagnosis, treatment recommendation, and information retrieval.
 - Managing ingestion, chunking, and embedding of over 200,000 PubMed documents to build a curated, searchable vector store optimized for biomedical retrieval.
 - Refactoring Radiant from a linear RAG+LLM pipeline into a modular, agent-based architecture using a mixture-of-experts framework.
 - Integrated a Partially Observable Markov Decision Process (POMDP) model to guide agent selection and optimize reward-based decision making.
- 2023–2024 **Project Assistant — Ramsey Lab**, Corvallis, OR
- Built the initial prototype of Radiant to support retrieval-augmented knowledge exploration in rare disease research.
 - Developed a document ingestion framework to extract biomedical literature from PubMed and transform it into structured inputs for downstream pipelines.
 - Evaluated multiple chunking strategies—including semantic segmentation, section-based splitting, and recursive character-based chunking—to optimize retrieval performance across heterogeneous biomedical texts.
 - Conducted comparative analyses of foundational LLMs versus fine-tuned variants to assess trade-offs in performance, hallucination risk, and relevance for rare disease literature.

Other Experience

- 2014–2020 **Ohio Army National Guard – Health Care Specialist/Combat Medic (68W)**
- 2020–2021 **Cleveland Clinic – Medical Assistant**
- 2021–2023 **FEMA – Contract EMT**

Independent Research Projects

- Stroke Prediction Pipeline Designed and implemented a machine learning pipeline for stroke prediction on a highly imbalanced clinical dataset. Evaluated models including Logistic Regression, Random Forest, and XGBoost. Applied preprocessing techniques such as SMOTE, PCA, and mean and nearest neighbor imputation. Achieved a recall of 0.835 and ROC AUC of 0.8279 with Random Forest (vs. 0.000 recall and 0.500 ROC AUC from a dummy classifier), demonstrating strong performance on the minority stroke class.

Leadership/Service/Teaching

- 2024–Present **Co-President – AI Graduate Student Association (AIGSA)**
Lead strategic planning initiatives, organize technical workshops and speaker events, and advocate for graduate student interests in artificial intelligence research and education.

- 2024–Present **Graduate Council Member – Faculty Senate**
Serve on the Graduate Council to review graduate program proposals, shape curriculum standards, and contribute to university-wide academic policy decisions.
- 2024–Present **Union Steward – Coalition of Graduate Employees (CGE)**
Represent graduate employees in contract negotiations and workplace concerns, facilitate communication between union leadership and department members, and uphold labor protections.
- 2023–2024 **Teaching Assistant – Data Structures & Computer Architecture/Assembly Language**
Instructed students through office hours, discussion sessions, and individual feedback; supported learning in algorithmic complexity, memory hierarchy, instruction sets, and low-level programming. Graded assignments and exams for 100+ students.

Publications

F.M. Hodges, et al., “Using AI to Improve Diagnosis and Treatment of Rare Diseases: A Chat Agent for Equitable and Accessible Healthcare,” *Artificial Intelligence in Medicine (AIME)*, LNCS vol. 15735, Springer, 2025. doi:10.1007/978-3-031-95841-0_35

F.M. Hodges, et al., White Paper on Radiant, Oregon State University, 2025. radiant.rtx.ai/whitepaper

Presentations

- Poster Presentation *Radiant: An Agentic RAG System for Rare Diseases*. Presented at Artificial Intelligence in Medicine, Pavia, Italy 2025.
- Demonstrative Presentation *Radiant*. Presented at Stanford & Research to the People Rare Disease Hackathon, San Francisco, California, 2024.